MEGHNA TIWARI

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Education

Lakshmi Narain College of Technology — Bhopal, India
B.Tech in Computer Science Engineering (IoT & Cyber Security including Blockchain Technology) — *2023 – 2027*

Carmel Convent Senior Secondary School — Bhopal, India
Higher Secondary Education — *2023*

Experience

Front-End Developer Intern — Sep 2025 – Nov 2025
IBM SkillsBuild
- Completed a Front-End Development internship, designing responsive UI components using **HTML**, **CSS**, and **JavaScript** with structured layouts, reusable components, and performance optimization.

GIS Intern — Aug 2025
Amay Environmental Consultancies — *Bhopal, India*
- Analyzed 10+ geospatial datasets using **QGIS**, improving reporting accuracy and compliance documentation.

Projects

Detect the Deceptive – Deepfake Detection Platform | *FastAPI, PyTorch, ConvNeXt, React, Docker*
- Built a deepfake detection system using **ConvNeXt** models trained in **PyTorch** to classify synthetic images and cloned audio.
- Evaluated performance across multiple datasets and applied **Grad-CAM** for model interpretability, achieving **85%+ validation accuracy**.
- Integrated **FastAPI**-based inference services and containerized deployment using **Docker**.

National Exam PYQ Test Platform (MERN Stack) | *React, Node.js, Express, MongoDB, JWT*
- Designed a full-stack examination platform with separate Admin and User dashboards using **React**, **Node.js**, and **Express**.
- Structured **MongoDB** schemas and developed secure **REST APIs** with **JWT-based authentication**, tested using **Postman**.
- Engineered a dynamic test interface with timer control, navigation logic, +4/-1 scoring, and automated result computation.

CyberGuard AI – Cyber Threat Intelligence Platform | *Flask, JavaScript, OSINT, Threat Modeling*
- Built a cybersecurity analysis tool using **OSINT** techniques to evaluate URLs, files, IP addresses, domains, and email headers.
- Implemented phishing detection with **SSL validation**, **SHA-256/MD5 hashing**, entropy scoring, and **SPF/DKIM/DMARC** verification.
- Generated unified **0–100 risk scores** and automated PDF threat reports for structured security assessment.

Autonomous UAV Survey Coverage System | *Python, A*, TSP, Computational Geometry*
- Designed a UAV coverage path planner using **polygon decomposition** and **Boustrophedon sweeping** for complete area coverage.
- Optimized cell visitation order using a **Traveling Salesman Problem -based strategy** and computed connector paths via **A* search** within polygon constraints.
- Simulated coverage missions in **matplotlib**, achieving **95%+ area coverage** with **25% reduced travel distance**.

Certifications

Microsoft: Azure AI Fundamentals (AI-900)
Cisco: Cybersecurity, Modern AI and Apply AI: Analyze Customer Reviews
NPTEL: Python Programming, Cryptography and Network Security, Blockchain, Cybersecurity, IoT and Industrial IoT
IBM: People and Soft Skills for Professional and Personal Success, Problem Solving with Creative and Critical Thinking
MathWorks: MATLAB, Simulink, Image Processing, Computer Vision, Stateflow
Coursera: People and Soft Skills Essentials

Skills

Programming Languages: Python, JavaScript (ES6+), HTML, CSS, MATLAB
Full Stack Development: React (Vite), Node.js, Express.js, Flask, RESTful APIs
Databases & Security: MongoDB, MongoDB Atlas, JWT Authentication, API Security
Software Engineering & Core CS: Software Development, Data Structures & Algorithms, OOP
Tools & Platforms: Git, GitHub, Postman, VS Code, QGIS, Google Earth Pro
Operating Systems & Professional Skills: Linux, Windows, Collaboration, Accountability, Communication, Problem Solving, Critical Thinking, Leadership